

## EDUCATION

### MSc Machine Learning, University College London

2023 – 2024

- First Class Honours (75.5%)
- Core Courses: Probabilistic and Unsupervised Learning (78%), Bayesian Deep Learning (83.5%), Supervised Learning (77.6%), Multi-agent Artificial Intelligence (85.5%), Information Retrieval and Data Mining (87.5%)

### BSc Mathematics, Imperial College London

2020 – 2023

- First Class Honours (80.3%)
- Core Courses: Methods for Data Science (94.4%), Mathematical Foundations of Machine Learning (90%), Probability for Statistics (80.7%), Scientific Computation (88.5%), Computational Linear Algebra (86.2%)

## EXPERIENCES

### Data Scientist, TESCO

Winter 2024

- Improved the product substitution model for recommending alternatives to out-of-stock items in online orders, and built and deployed data (ETL) pipelines with PySpark, **delivering £3M+ in annual business savings**
- Led the PoC phase of an AI recipe recommendation system, by developing an LLM pipeline to automatically construct a recipe-to-product knowledge graph, powering a **RAG-based chatbot** for personalized recipe suggestions
- Designed an **LLM-as-a-Judge evaluation framework** with Langfuse to automate the assessment of AI-generated content, calibrating its outputs against human benchmarks to ensure alignment with qualitative standards at scale
- Achieved **top 5 performer** out of the 200+ member Data Science team in the second quarter after joining

### Off-Cycle Intern, Global Banking & Markets Department of Goldman Sachs

Autumn 2024

- Developed and deployed a market report automation in Slang to support trader decision-making, integrating data pipelines and formatting logic to enable daily email delivery of structured insights that were previously unavailable
- Initiated a React-based internal dashboard to transition the team from static email reports to web-based analytics

### Intern, Engineering Department of Goldman Sachs

Spring 2021

- Built and presented a Python-based model for hedging natural gas futures, analyzing five years of market and weather data; used neural networks for price prediction and Value-at-Risk to optimize trade quantities
- Developed a strategy that satisfied order book, cash, and daily loss limits, ranked **1st out of 10 teams**

## PROJECTS

### Spreadsheet Copilot Agent

Summer 2025

- Built a Spreadsheet Copilot using a **ReAct LLM agent with self-reflection**, capable of safely generating and executing code in a secure sandbox to resolve complex spreadsheet tasks in real time
- Enhanced performance by fine-tuning (LoRA) the underlying Qwen2.5-Coder-7B-Instruct model with high-quality synthetic CoT data (using Kimi-K2-Instruct), simulating diverse tasks and user personas to improve generalization

### Large Language Models for Product Substitution, UCL & TESCO

Summer 2024

- Transformed product substitution into a solvable NLP challenge, using LLMs (GPT-4o and Llama 3) with a **RAG-based, augment-then-summarise approach** to inject model knowledge into product metadata
- Prepared and processed a historical user behavior dataset to **fine-tune a cross-encoder model** (ms-marco-MiniLM-L6-v2) with binary cross entropy loss, improving the re-ranking of candidate substitutions
- Demonstrated significant performance gains over baselines in experiments, achieving **+0.73% NDCG** and **+32.51% NDCG<sub>f</sub>**, validating the effectiveness of **productionising research** for a real-world business problem

### Blockchain Coder: Enhancing Code Generation with Multi-Agent Framework, UCL

Summer 2024

- **Designed a multi-agent framework** (Planner, Optimizer, Debugger) to improve true generalisation over simple memorization, addressing the ~40% accuracy drop that LLMs exhibit on previously unseen coding problems
- Deconstructed complex problems and refined code via unit-test-driven and question-based iteration, **doubling the pass@1 success rate** (to 6.90%) and increasing the private test case pass rate by **5.7x** (to 38.08%)

## TECHNICAL SKILLS

|              |                                                                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages    | Python, SQL, R, Julia, Mandarin (Native), Cantonese (Native)                                                                                           |
| Technologies | PyTorch, TensorFlow, HuggingFace, LangChain, AutoGen, vLLM, Milvus, Langfuse, PySpark, Pandas, Airflow, Docker, Git, RAG, Fine-tuning (SFT, PEFT, DPO) |

## ACHIEVEMENTS

- **ElevenLabs × LFH × Bolt Hackathon** ranked 2/50, created Sign2Speak an AI sign language translation system
- **AI Content Creator MarketVision**, an open-source AI stock tracker, gained 1K+ followers through one tutorial
- **LLM Post-training Course**, NexusFlow × University of Washington, hands-on with SFT, DPO, and Online RL